



Objective:

This project focuses on organizing, analyzing, and extracting insights from a product dataset. Using fundamental Excel functions, the workflow transforms raw product metrics such as handling missing values, correcting inconsistencies, removing duplicate records, restructuring columns, and applying formatting rules to improve the usability and readability of the dataset.

Data set Description:

This data set contains detailed retail information for a variety of Electronic products. It tracks core product attributes, pricing structures, and current stock volumes. The data is used to analyze inventory levels, categorize items and monitor overall inventory health.

Key Data Included:

Product Details: Basic identification including names and categories.

Pricing Metrics: Individual product price

Inventory: Individual product quantity.

Column Description:

Column Name	Description
Product ID	ID of the product
Product Name	Name of the product
Brand Name	It shows brand name of the product
Price	It shows price of the product
Quantity	It shows the quantity of product we have.
Category	It shows the product is basically from which categories.

Data Exploration:

We have done the data exploration using excel formula's as per the requirements.

The Requirements and Outcomes are,

1. Handling Missing Values:

a. Missing Price values:

- Finding missing values in “Price” column by using ISBLANK()
- And add average of total price values
- Using IF() search blank and adding average value on it.

Outcomes:

=IF(ISBLANK(F7),AVERAGE(\$F\$2:\$F\$30),F7)					
	C	D	E	F	G
nt	Product Nam	Brand Nar	Price (Handling M		
	Product Nam	Brand Nar	issing Values)	Price (\$)	Quan
	laptop	Dell	\$1,000	1000	30
	Sneakers	Nike	\$80	80	15
	Coffee Maker	Keurig	\$130	130	40
	smartphone	Samsung	\$900	900	25
	Backpack	North Face	\$70	70	20
	Headphones	Sony	\$284		45
	T-shirt	Adidas	\$30	30	5
	Blender	Ninja	\$90	90	35
	Tablet	Apple	\$500	500	50

b. Missing category:

- Finding missing values in “Category” column by using ISBLANK()
- Using IF() search blank if it is blank display as “Unknown” on it.

Outcomes

=IF(ISBLANK(I6),"Unknown",I6)

	H	I	J
ent	Category (Handling Missing Values)	Category	
	Electronics	Electronics	
	Fashion	Fashion	
	Kitchen	Kitchen	
	Electronics	Electronics	
	Unknown		
	Electronics	Electronics	
	Fashion	Fashion	
	Kitchen	Kitchen	
	Electronics	Electronics	
	Outdoor	Outdoor	

2. Correcting Inconsistent Data:

a. Correcting Data:

- Finding inconsistent data in “Product Name” column using CLEAN(), TRIM(), PROPER() correcting the data.

Outcomes:

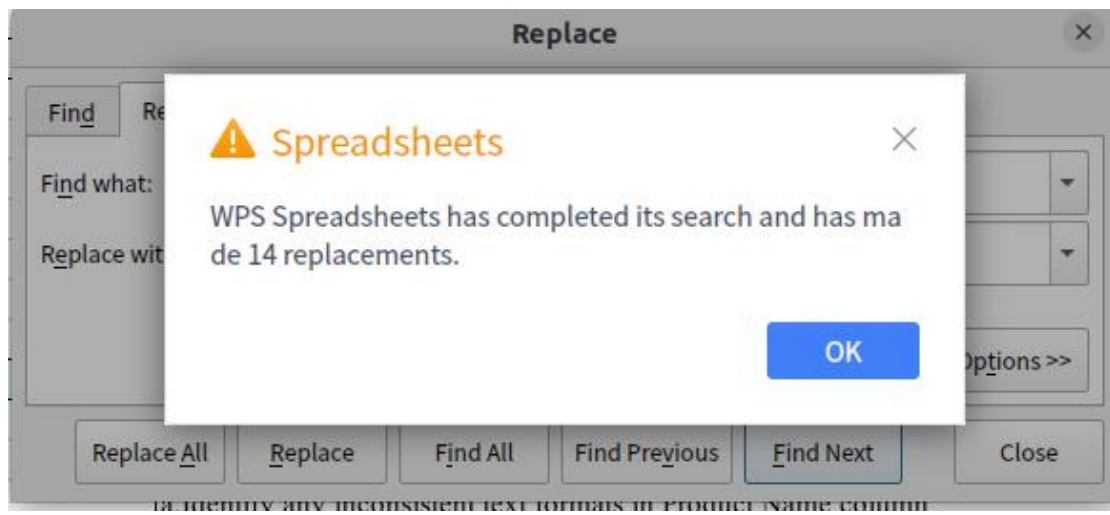
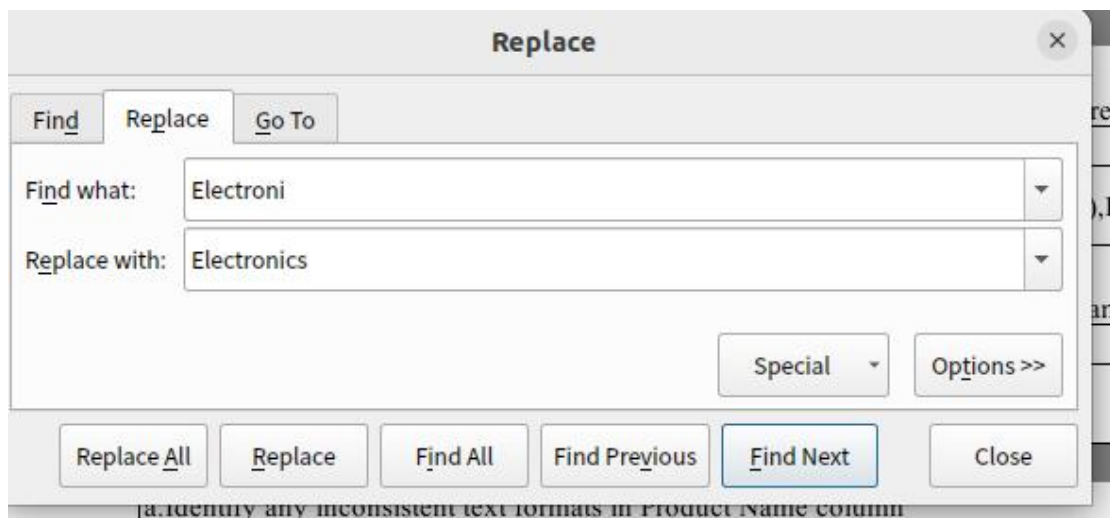
=PROPER(C2)

	B	C
	Correcting inconsistent Data (Product Name)	Product Name
	Laptop	laptop
	Sneakers	Sneakers
S	Coffee Maker	Coffee Maker
	Smartphone	smartphone
S	Backpack	Backpack
	Headphones	Headphones
	T-Shirt	T-shirt
K	Blender	Blender

b. Standardize text & Misspelling :

- Finding the Misspelling in “Category” column & standardized the text format by using Find & replace function.

Outcomes:



3. Removing Duplicate:

- Finding the duplicate by using Data Tab ➡ Remove Duplicate
- Click on “Remove duplicate “ function duplicates are removed.

Outcomes

17-JUN-IN	Laptop	Laptop	HP	950	950	25	Electronics	Electronics
16-APR-ES	Headphones	headphones	Bose	250	250	20	Electronics	Electronics
21-AUG-CA	Laptop Bag	Laptop Bag	Samsonite	50	50	35	Accessories	Accessories
21-AUG-CA	Laptop Bag	Laptop Bag	Samsonite	50	50	35	Accessories	Accessories
16-APR-ES	Headphones	headphones	Bose	250	250	20	Electronics	Electronics
17-JUN-IN	Laptop	Laptop	HP	950	950	25	Electronics	Electronics

4. Splitting & Merging Data:

a. Splitting Data:

- By Using LEFT(), RIGHT() Splitting the “Product ID” column into “Manufacturing Date” column and “Country Code”.

Outcomes:

The first screenshot shows the formula `=LEFT(A2,6)` in the formula bar. The spreadsheet has two columns: B, labeled 'Manufacturing Date', and D, labeled 'Product Brand'. The value '28-JAN' is in column B and 'Laptop - Dell' is in column D.

The second screenshot shows the formula `=RIGHT(A2,2)` in the formula bar. The spreadsheet has two columns: C, labeled 'Country Code', and D, labeled 'Product Brand'. The value 'US' is in column C and 'Laptop - Dell' is in column D.

b. Merging Data:

- Merge the “Brand Name” and “Product Name” columns into one column named “Product Brand” by using “&”.

Outcomes

The screenshot shows the formula `=E2&" - "&G2` in the formula bar. The spreadsheet has a single column D, labeled 'Product Brand'. The data in this column is as follows:

Product Brand
Laptop - Dell
Sneakers - Nike
Coffee Maker - Keurig
Smartphone - Samsung
Backpack - North Face
Headphones - Sony
T-Shirt - Adidas
Blender - Ninia

5. Number Formatting:

a. Formatting Data type in “Price” column:

- Formatting the data type of “Price” column using Home tab formatting function.

Outcomes:

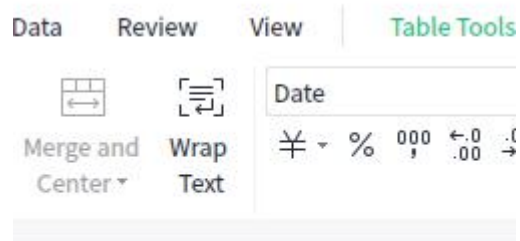


H	I	J
Brand Name	Price (Handling)	Price
Bell	\$1,000.00	1000
Like	\$80.00	80

b. Formatting Data type in “Manufacturing Date” column:

- Formatting the data type of “Manufacturing Date” column using Home tab formatting function.

Outcomes



	H	I	J
	Manufacturing Date	Price (\$)	
	28-JAN	1000	30
	15-FEB	80	15
	03-MAR	130	40
ng	11-APR	900	25
Face	22-MAY	70	20
	07-JUN		45
	19-JUL	30	5
	23-AUG	90	35
	05-SEP	500	50

6. Conditional Formatting:

a. Apply Data Bar Or Color Scale:

- Using Home Tab - Conditional Formatting function we applying data bar for the price column.

Outcomes:

H
Price (Handling Missing)
\$1,000.00
\$80.00
\$130.00
\$900.00
\$70.00
\$283.60
\$30.00
\$90.00
\$500.00
\$130.00
\$90.00
\$120.00
\$150.00
\$160.00

b. Highlights the cell contains “Electronics”:

- Using Home Tab - Conditional Formatting - new rule we highlights the cells where the “category” is “Electronics”.

Outcomes:

K
Category (Handling Missing)
Electronics
Fashion
Kitchen
Electronics
Unknown
Electronics
Fashion
Kitchen
Electronics
Outdoor
Unknown
Unknown
Electronics
Electronics
Electronics

CONCLUSION:

By applying fundamental Excel formulas to the raw dataset, the project requirements were successfully met. The formulas successfully converted chaotic product metrics into a structured, actionable summary of inventory health and performance. The Overall outcomes are,

Product ID	Manufacturing Date	Country Code	Product Brand	Inconsistent Data	Product Name	Brand Name	(Handling Missing)	Price (\$)	Quantity	(Handling Missing)	Category
28-JAN-US	28-JAN	US	Laptop - Dell	Laptop	laptop	Dell	\$1,000.00	1000	30	Electronics	Electronics
15-FEB-US	15-FEB	US	Sneakers - Nike	Sneakers	Sneakers	Nike	\$80.00	80	15	Fashion	Fashion
03-MAR-US	03-MAR	US	Coffee Maker - Keurig	Coffee Maker	Coffee Maker	Keurig	\$130.00	130	40	Kitchen	Kitchen
11-APR-US	11-APR	US	Smartphone - Samsung	Smartphone	smartphone	Samsung	\$900.00	900	25	Electronics	Electronics
22-MAY-US	22-MAY	US	Backpack - North Face	Backpack	Backpack	North Face	\$70.00	70	20	Unknown	
07-JUN-UK	07-JUN	UK	Headphones - Sony	Headphones	Headphones	Sony	\$283.60		45	Electronics	Electronics
19-JUL-UK	19-JUL	UK	T-Shirt - Adidas	T-Shirt	T-shirt	Adidas	\$30.00	30	5	Fashion	Fashion
23-AUG-UK	23-AUG	UK	Blender - Ninja	Blender	Blender	Ninja	\$90.00	90	35	Kitchen	Kitchen
05-SEP-UK	05-SEP	UK	Tablet - Apple	Tablet	Tablet	Apple	\$500.00	500	50	Electronics	Electronics
14-OCT-UK	14-OCT	UK	Hiking Boots - Timberland	Hiking Boots	Hiking Boots	Timberland	\$130.00	130	10	Outdoor	Outdoor
25-NOV-AU	25-NOV	AU	Sneakers - Adidas	Sneakers	Sneakers	Adidas	\$90.00	90	40	Unknown	
08-DEC-DE	08-DEC	DE	Coffee Maker - Nespresso	Coffee Maker	Coffee Maker	Nespresso	\$120.00	120	35	Unknown	
18-FEB-CA	18-FEB	CA	Smartwatch - Fitbit	Smartwatch	Smartwatch	Fitbit	\$150.00	150	15	Electronics	Electronics
20-AUG-CN	20-AUG	CN	Smartwatch - Huawei	Smartwatch	Smartwatch	Huawei	\$160.00	160	15	Electronics	Electronics
27-JAN-JT	27-JAN	JT	Laptop - Asus	Laptop	laptop	Asus	\$980.00	980	10	Electronics	Electronics
01-MAR-UK	01-MAR	UK	Sunglasses - Oakley	Sunglasses	Sunglasses	Oakley	\$150.00	150	15	Fashion	Fashion
14-AUG-US	14-AUG	US	Camping Tent - Coleman	Camping Tent	Camping Tent	Coleman	\$283.60		10	Outdoor	Outdoor
14-MAY-RU	14-MAY	RU	Camera - Nikon	Camera	Camera	Nikon	\$700.00	700	50	Electronics	Electronics
09-JAN-CA	09-JAN	CA	Microwave - Panasonic	Microwave	Microwave	Panasonic	\$80.00	80	20	Kitchen	Kitchen
19-JUL-BR	19-JUL	BR	Fitness Tracker - Xiaomi	Fitness Tracker	Fitness Tracker	Xiaomi	\$150.00	150	30	Unknown	
29-SEP-CA	29-SEP	CA	Smartphone - Google	Smartphone	Smartphone	Google	\$800.00	800	45	Electronics	Electronics
03-JUN-CA	03-JUN	CA	Sunglasses - Ray-Ban	Sunglasses	Sunglasses	Ray-Ban	\$283.60		25	Fashion	Fashion
11-JUL-CA	11-JUL	CA	Blender - Vitamix	Blender	Blender	Vitamix	\$400.00	400	40	Kitchen	Kitchen
07-MAR-CA	07-MAR	CA	Dress - Zara	Dress	Dress	Zara	\$60.00	60	30	Fashion	Fashion
13-APR-CA	13-APR	CA	Toaster - Hamilton	Toaster	Toaster	Hamilton	\$40.00	40	10	Kitchen	Kitchen
24-MAY-CA	24-MAY	CA	Fitness Tracker - Garmin	Fitness Tracker	Fitness Tracker	Garmin	\$130.00	130	5	Electronics	Electronics
02-DEC-CA	02-DEC	CA	Jeans - Levi's	Jeans	Jeans	Levi's	\$50.00	50	50	Fashion	Fashion
09-JUL-FR	09-JUL	FR	Watch - Casio	Watch	Watch	Casio	\$100.00	100	20	Accessories	Accessories

1.Handling Missing Values			
*A.Missing Price			
Using Conditional Formatting IF() function- Find the blank in Price column and replace average of the total price			
Formula	Function	Result	Detailed Description -Example of row
IF(), ISBLANK(), AVERAGE()	IF(ISBLANK(E7),AVERAGE(\$E\$2:\$E\$35),E2)	₹1,000.00	Found blank in 7th row of price column and replace average of total price by using IF()
*B.Missing Categories			
Using Conditional Formatting IF() function- Find the blank in Category column and replace as "Unknown"			
Formula	Function	Result	Detailed Description -Example of row
IF(), ISBLANK()	IF(ISBLANK(I2),"UnKnown",I2)	Unknwn	Found blank in 6th row of Category column and replace as Unknown by using IF(), ISBLANK()
2. Correcting inconsistent Data			
a.Identify any inconsistent text formats in Product Name column			
Found inconsistent data by using Proper() correcting the text format			
Formula	Function	Result	Detailed Description -Example of row
Proper()	PROPER(C2)	Laptop	Found inconstant Data in 2th Row as laptop by using proper() convert standardize text formats
b.Identify Typos & Misspelling in Category column			
Found Misspelling Using filterer & Using find and replace function fixing it.			
Formula	Function	Result	Detailed Description -Example of row
Filler, Find & Replace	Crtl + H (Or) In Home Tab we choose Find & replace Option	Electronics	Found Electroni in 7th row by using find & replce function changed as Electronics
3.Removing Duplicates			
Finding duplicate value in Product ID column			
Formula	Function	Result	Detailed Description -Example of row
Choose Remove Duplicate in Data Tab	Select Product ID column & Remove Duplicate in Data Tab	Duplicate Data removed	Found 6 rows of duplicate items & Removed.

4. Splitting & Merging Data		
a. Splitting Product ID column into Two as "Manufacturing Date" & "Country Code"		
Using LEFT(),RIGHT() returned Manufacturing Date, Country Code in two Seperate Column		
Formula	Function	Results
LEFT(), RIGHT()	LEFT(A2,6), RIGHT(A2,2)	28-JAN US
b.Merging Product Name and Product Brand column into "Product Brand" column		
Merging Data by using "&"		
Formula	Function	Results
&	F2&" - "&H2	Laptop - Dell
5. Number Formatting		
Format the data type of Price column into Currency & Manufacturing Date into Date type		
Formula	Function	Results
Home Tab	Home Tab - change Currency, Date Format	Currency, Date format changed
6.Conditional Formatting		
a.Apply Data bar or Color scales formatting in the price column		
Formula	Function	Results
Home Tab	Home Tab - Conditional formatting using Data Bar	Data Bar Shown in the column
b.Custom rule for conditonal formatting in the category column to highlight Electronics cell		
Formula	Function	Results
Home Tab	Home Tab - Conditional formatting using highlight which contains " Electronics "	In category column "Electronics" Highlights in purple color